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SmartStay Hostel Management System — UML Class Diagram

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1. UML Notation Legend

Access Modifiers

Symbol	Meaning
+	**Public** — accessible from anywhere
−	**Private** — accessible only within the class
#	**Protected** — accessible within the class and subclasses
~	**Package** — accessible within the same module
$	**Static** — belongs to the class itself, not instances

Relationships & Arrows

Mermaid Syntax	UML Name	Visual	Description
`<|--`	**Inheritance**	——▷	Solid line + closed triangle. Class extends another.
`<|..`	**Realization**	----▷	Dashed line + closed triangle. Class implements an interface.
`-->`	**Directed Association**	——▶	Solid line + open arrow. A stores B as a field (one-way).
`o--`	**Aggregation**	◇——	Solid line + open diamond. "Has-a" — parts exist independently.
`*--`	**Composition**	◆——	Solid line + filled diamond. "Owns" — parts die with the whole.
`..>`	**Dependency**	----▶	Dashed line + open arrow. Temporary use (not stored as field).

Dependency Stereotypes

Label	Meaning	Example
`«use»`	Calls methods on B temporarily	`Repository «use» PrismaClient`
`«create»`	Creates new instances via `new B()`	`PaymentService «create» CreditCardStrategy`

> [!NOTE] `«use»` and `«create»` are **NOT** separate relationship types — they are **labels on the same Dependency arrow** (`..>`) to clarify the nature of the dependency.

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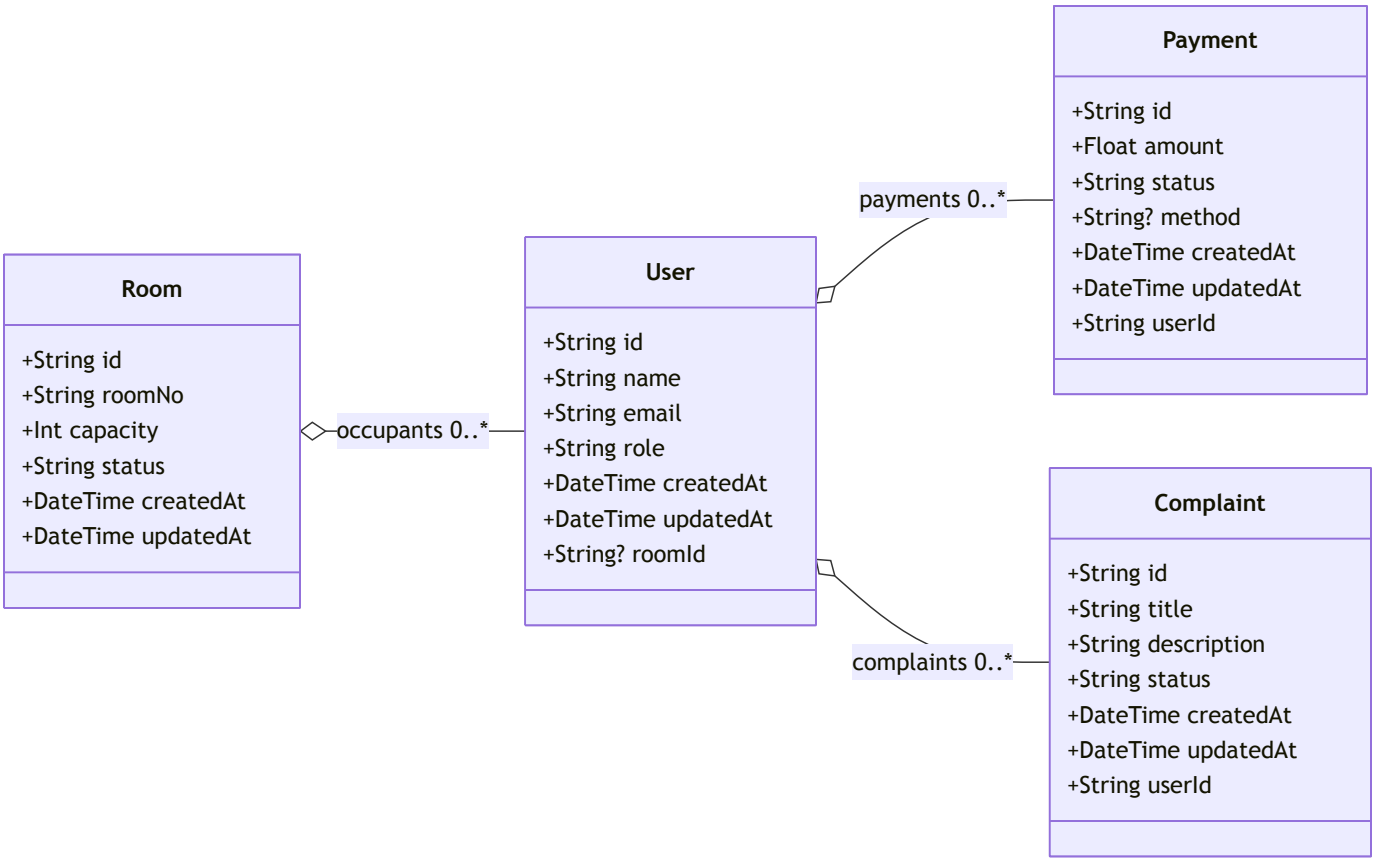
2. Domain Models

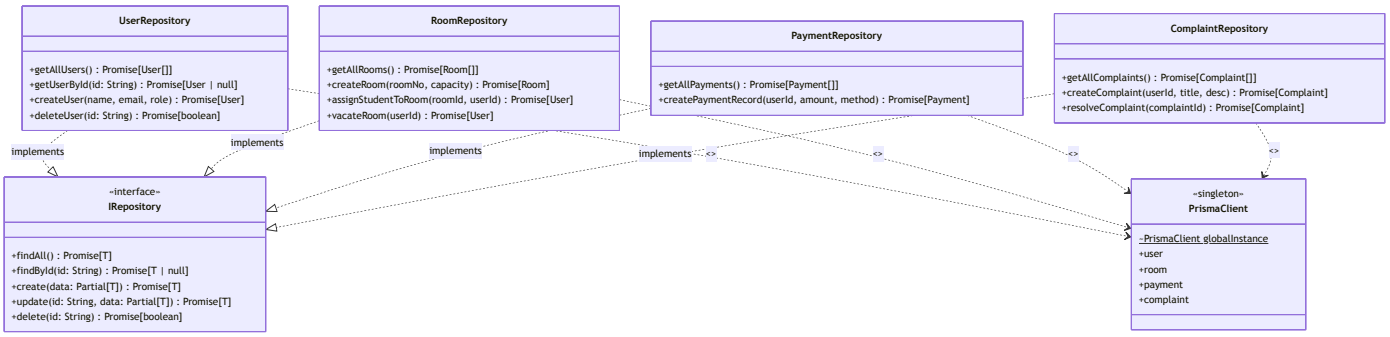
> Entities derived from the Prisma schema — shows data structure and how models relate to each other.



Relationship	Type	Why?
`User` ⟶ `Room`	**Directed Association**	User stores `roomId` — User knows its Room, but Room doesn't store a userId.
`Room` ◇— `User`	**Aggregation**	Room has occupants, but Users can exist without a Room.
`User` ◇— `Payment`	**Aggregation**	User has Payments, but Payments can be queried independently.
`User` ◇— `Complaint`	**Aggregation**	User has Complaints, but Complaints can be queried independently.

3. Repository Layer

> Abstraction over data access. All concrete repositories realize the generic `IRepository<T>` interface and use the PrismaClient singleton via `«use»` dependency.



Relationship	Type	Why?
`*Repository` ⇢▷ `IRepository`	**Realization**	Concrete repos implement the generic interface contract.
`*Repository` ⇢▶ `PrismaClient`	**Dependency «use»**	Repos call `prisma.model.method()` — they use the singleton but don't store it as a field.

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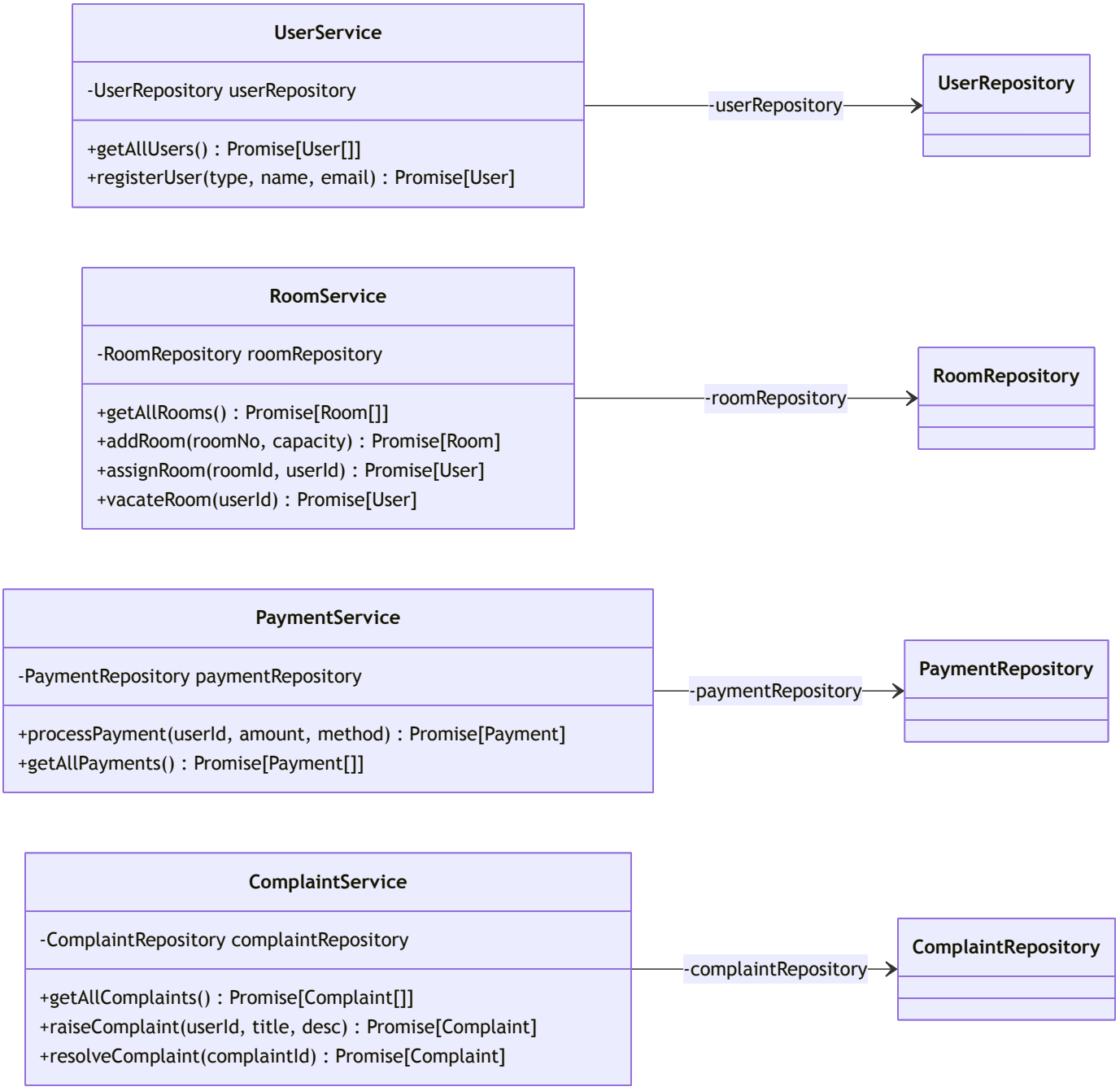
4. Service Layer

> Business logic layer. Each service holds a **private** reference to its repository (Directed Association) and delegates data operations to it.



Relationship	Type	Why?
`Service` ⟶ `Repository`	**Directed Association**	Each service stores its repository as a**private field** (–). One-way: Service knows Repository, Repository does NOT know Service.

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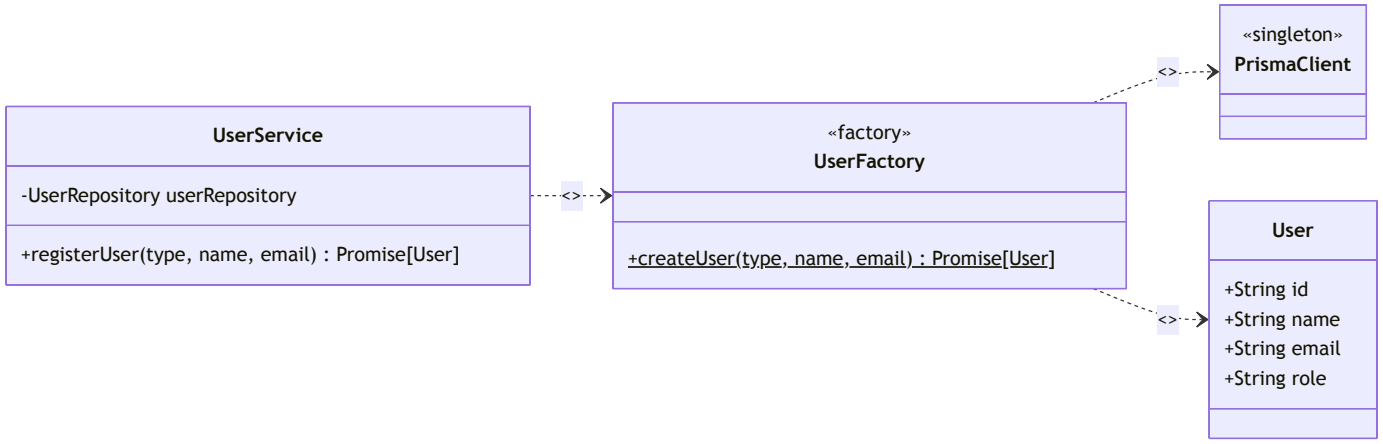
5. Factory Pattern — UserFactory

> Encapsulates User creation logic. The `UserService` depends on `UserFactory` via a `«use»` dependency (calls its static method, does not store it).



Relationship	Type	Why?
`UserService` ----▶ `UserFactory`	**Dependency «use»**	Service calls `UserFactory.createUser()` — a static method call, no field stored.
`UserFactory` ----▶ `PrismaClient`	**Dependency «use»**	Factory calls `prisma.user.create()` internally.
`UserFactory` ----▶ `User`	**Dependency «create»**	Factory instantiates User records.

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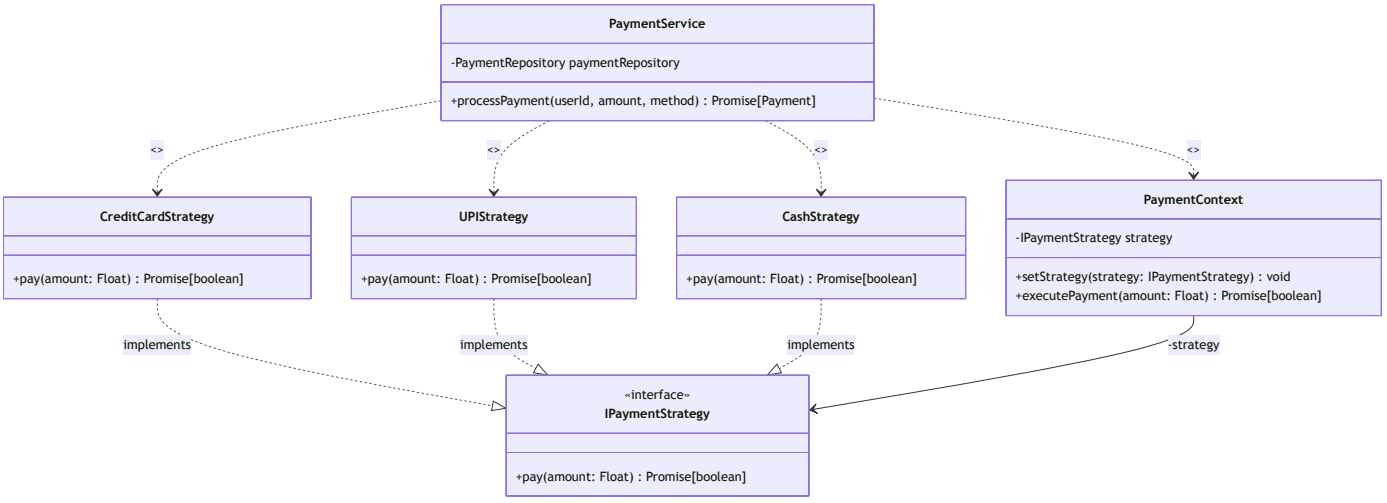
6. Strategy Pattern — Payment Processing

> Allows swapping payment algorithms at runtime. `PaymentService` creates (`«create»`) the context and concrete strategies. `PaymentContext` stores the strategy as a private field (Directed Association).



Relationship	Type	Why?
`*Strategy` ⇢ `IPaymentStrategy`	**Realization**	Concrete strategies implement the interface.
`PaymentContext` ⟶ `IPaymentStrategy`	**Directed Association**	Context stores active strategy as a**private field** (`-strategy`).
`PaymentService` ⇢ `*Strategy`	**Dependency «create»**	Service does `new CreditCardStrategy()` etc. — creates instances but doesn't store them.
`PaymentService` ⇢ `PaymentContext`	**Dependency «create»**	Service does `new PaymentContext(strategy)` — creates it locally per request.

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7. Full System Overview (Simplified)

> High-level view showing all layers and how they connect. Classes are simplified to show only the relationship structure.

```
Parse error on line 4:
...    %% Domain Models CLASS User { }  cl
----------------------^
Expecting 'CLASS', 'MEMBER', 'SEPARATOR', 'UNICODE_TEXT',
'NUM', 'ALPHA', got 'NEWLINE'
```

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8. Summary Tables

All Access Modifiers Used

Modifier	Symbol	Where Used
Public	`+`	All domain attributes, all repository & service public methods, strategy `pay()`, factory `createUser()$`

Modifier	Symbol	Where Used
Private	`-`	`UserService.userRepository`, `RoomService.roomRepository`, `PaymentService.paymentRepository`, `ComplaintService.complaintRepository`, `PaymentContext.strategy`
Package	`~`	`PrismaClient.globalInstance` (module-scoped singleton)
Static	`$`	`UserFactory.createUser()`

All Relationship Types Used

#	Type	Arrow	Count	Where
1	**Aggregation**	◇——	3	Room↔User, User↔Payment, User↔Complaint
2	**Directed Association**	——▶	6	User→Room, 4× Service→Repository, Context→Strategy
3	**Realization**	----▷	7	4× Repository→IRepository, 3× Strategy→IPaymentStrategy
4	**Dependency «use»**	----▶	6	4× Repository→Prisma, Factory→Prisma, Service→Factory
5	**Dependency «create»**	----▶	5	PaymentService→Context, PaymentService→3 Strategies, Factory→User

Design Patterns

Pattern	Where	Key Classes
Singleton	`src/lib/prisma.ts`	`PrismaClient`
Factory Method	`src/factories/`	`UserFactory` → `User`

Pattern	Where	Key Classes
Strategy	`src/strategies/`	`IPaymentStrategy`, `CreditCardStrategy`, `UPIStrategy`, `CashStrategy`, `PaymentContext`
Repository	`src/repositories/`	`IRepository<T>`, `UserRepository`, `RoomRepository`, `PaymentRepository`, `ComplaintRepository`